

Analysis Report

Methodology and Approach:

The aim was to forecast the cancellations of hotel booking (is_canceled). The data was heavily cleaned, that is, it was cleaned to deal with blank values in the agent, company, and country columns. The major issue was Data Leakage as the columns such as reservation status would only be updated once the cancellation has taken place and this would result in an over-optimistic model.

The data was divided into a 70% Training set to be trained on the patterns, a 15% Validation set to be fine-tuned on the pattern of model selection, a 15% Test set to provide a fair final assessment. The processing of features was performed in a pipeline that was able to process numeric scaling and one-hot encoding concurrently.

Model Performance & Discussion:

Best Individual Model: Gradient Boosting was the best performing standalone model. It scored an F1-score of 0.788. The fact that it was able to correct mistakes made by the previous trees allowed the tree to capture complex interactions between lead-time, deposit-type, and total-of-special-requests.

The Observation, which is the "SVC3" Observation: The Support Vector Classifier with a degree-3 polynomial kernel was performing fairly well (Acc: 82.5%) but was computationally expensive and failed to perform as well as the tree-based models on this tabular data.

Ensemble Analysis: The Bayesian Stacking Ensemble (with Gradient Boosting, Random Forest, and Decision Tree as the base learners and a Naive Bayes meta-learner) was the overall winner.

Why? Although the Voting Classifier merely averages predictions, the Bayesian Stacker uses the meta-learner to "learn" what model to rely on when handling particular kinds of data points. With the final layer being an Naive Bayes estimator, it effectively estimates the posterior probability of a cancellation given the opinions of the base models, which also results in the highest Recall (0.738) and F1-score (0.800).

Conclusion:

In the case of a hotel, a prediction of a cancellation being missed (False Negative) will lead to an empty room and missed revenue. As such the Bayesian Stacking Ensemble is a better one to use in production because it is better at identifying any actual cancellations and yet has a high overall accuracy.